

Density-Based Dynamic Ensemble Learning with Gaussian Mixture Model for Enhanced Predictive Analytics.

Abstract—Dynamic ensemble learning is a powerful technique for enhancing predictive accuracy and robustness by combining multiple classifiers. However, traditional methods face challenges in dynamic environments, including difficulty in providing personalized predictions, sensitivity to label noise, and the need to balance classifier diversity with ensemble generalization. This paper introduces a novel Density-Based Dynamic Ensemble Learning Algorithm using Gaussian Mixture Models to address these issues. The proposed algorithm leverages Gaussian Mixture Models to capture complex data structures, creating subsets that reflect specific patterns, which are then used to train distinct base classifiers, ensuring diversity within the ensemble. The density-based dynamic selection strategy intelligently selects classifiers for each test instance based on its proximity to these clustered subsets, enhancing accuracy and adaptability. The algorithm is evaluated against single ensemble methods as well as other dynamic ensemble methods to assess its performance and effectiveness, using logistic regression, support vector machine, k-nearest neighbors, and decision tree as base learners. Results demonstrate that the proposed method, particularly when combined with logistic regression and support vector machine, outperforms other methods, underscoring the effectiveness of the density-based selection strategy. Further analysis shows that increasing the number of clusters and sampling ratio improves the F-score, though with diminishing returns beyond a certain point. These findings highlight the importance of base learner selection and parameter tuning in achieving robust and adaptable ensemble performance, positioning the proposed algorithm as a significant advancement in handling complex, high-dimensional data.

Index Terms—Dynamic Ensemble Learning, Gaussian Mixture Models, Density-Based Ensemble Learning, Label Noise Sensitivity, Predictive Analytics, Model Personalization, Decision-Making Efficiency.

I. Introduction

THE advent of dynamic ensemble classifiers heralds a new epoch in machine learning, addressing critical needs across sectors such as medical diagnostics [1], cybersecurity [2], and more. Recent advancements have shown that these classifiers significantly enhance the accuracy and robustness of human activity recognition (HAR) systems, as well as medical diagnostics, particularly in fields like cardiovascular disease detection using novel ensemble classifiers [3]. For instance, evolutionary dual-ensemble class imbalance learning methods have demonstrated superior performance in recognizing diverse human actions by optimizing the combination of base classifiers [4]. Ensemble learning, by synergizing multiple classifiers, offers unparalleled robustness and accuracy, far surpassing the capabilities of individual classifiers [5]–[9]. This approach, known as Multiple Classifier Systems (MCS),

leverages diverse models for enhanced prediction reliability. Specifically, Dynamic Ensemble Selection (DES) refines this concept by customizing classifier combinations for each test instance, thus ensuring adaptability and precision in predictive analytics [10], [11]. The sophistication of ensemble methods is evident in their application to complex tasks, where they have consistently outperformed single-model approaches [12], [13]. Despite the advancements, ensemble learning faces significant challenges. Key among these is the need for personalized predictions in dynamic environments, such as fluctuating market trends or evolving cybersecurity threats [14]. Traditional ensemble methods struggle with the nuanced balance between classifier diversity and ensemble generalization, a barrier to harnessing their full potential [15]. The influence of label noise on classification performance is another major concern, leading to potential overfitting and reduced efficacy in practical applications. Furthermore, the dilemma of maintaining robustness without increasing ensemble complexity presents a persistent challenge in dynamic selection methods. These issues underscore the necessity for innovative solutions that can adapt to variable data conditions while maintaining accuracy and efficiency [14].

Our proposed model, Density-Based Dynamic Ensemble Learning Using Algorithm Gaussian Mixture Model (DDELA-GMM), addresses these challenges with a groundbreaking approach in dynamic ensemble learning. By integrating Gaussian Mixture Models (GMM) for sophisticated data clustering and a density-based dynamic selection strategy, DDELA-GMM enhances personalization and accuracy in prediction tasks. Specifically, it is designed to tackle the challenges of high-dimensional spaces and label noise sensitivity, which are common hurdles in traditional ensemble methods. By advancing beyond these limitations, DDELA-GMM sets a new standard for adaptability, precision, and decision-making efficiency in the field. This model does not merely improve existing methods; it represents a paradigm shift in how ensemble learning can be approached, promising significant advancements in various complex and dynamic scenarios.

This study leverages a complex and high-dimensional dataset generated from wearable sensor technology to rigorously evaluate the performance of the proposed dynamic ensemble learning model. The dataset originates from recordings of human activity, captured using accelerometers and gyroscopes integrated into smartphones. These sensors meticulously track the subtle movements and changes in orientation as subjects engage in various

physical activities, providing a rich source of information for activity recognition tasks.

Each recording in the dataset corresponds to a distinct activity performed by a subject and encapsulates a time series of sensor readings. At each time step within a recording, the accelerometers and gyroscopes measure linear acceleration and angular velocity, respectively, across three orthogonal axes (X, Y, and Z). This intricate sampling process generates a high-dimensional feature space, where each feature represents the magnitude of a specific sensor measurement (acceleration or angular velocity) along a particular axis at a given time instance.

The raw sensor data, while rich in information, is inherently prone to noise and uncertainty. This stems from a confluence of factors inherent to the nature of wearable sensing:

- **Sensor Imperfections:** Like all physical instruments, accelerometers and gyroscopes exhibit limitations in their sensitivity and accuracy. Minute variations in sensor readings can arise from internal electronic noise, temperature fluctuations, and manufacturing inconsistencies.
- **Environmental Interference:** The surrounding environment can introduce spurious signals that contaminate the sensor readings. Electromagnetic interference from nearby electronic devices, vibrations from the floor or other surfaces, and even changes in air pressure can all contribute to noise in the data.
- **Placement and Movement Variability:** The placement of the smartphone on the subject's body (e.g., pocket, belt, hand) and variations in the manner of movement can significantly influence the sensor readings. Slight shifts in the device's position or changes in the execution of an activity introduce inconsistencies into the collected data.

These inherent challenges of high dimensionality, noise, and uncertainty demand a sophisticated classification approach. This is where the proposed dynamic ensemble model demonstrates its strength. By employing Gaussian Mixture Models (GMMs), the model effectively captures the underlying data distribution, accounting for its multimodality and effectively distinguishing meaningful patterns from random fluctuations. Moreover, the model's density-based selection strategy ensures that for each test instance, the most appropriate base learners, trained on relevant data partitions, are selected for prediction. This dynamic and context-aware approach enhances the model's robustness and accuracy, enabling it to achieve high performance even when faced with the complexities of real-world sensor data.

II. Literature Review

A. Ensemble Learning and Dynamic Ensemble Selection

Ensemble learning marks a significant stride in machine learning, particularly in the context of predictive modeling. Combining multiple models, such as through Bagging, Random Subspace Method, and Random Forest, ensemble

learning approaches have improved predictive accuracy and robustness [16]. The diversification among base models is key to the success of these methods. Furthermore, dynamic ensemble selection techniques have evolved, allowing for the adaptation of model selection based on each test sample's specific characteristics, thereby offering more personalized predictions [17]. This is particularly useful in applications like breast cancer detection, where an ensemble of deep learning networks has been shown to improve diagnostic accuracy using consensus-adaptive weighting techniques [18]. The DeepCluE approach employs a convolutional neural network to generate diverse clusterings from multiple layers, effectively selecting the most informative ones for final clustering [19]. Similarly, the probabilistic combination of non-linear eigenprojections for ensemble classification enhances the reliability of classifications by integrating a probabilistic support vector machine (SVM) approach, allowing for better handling of uncertainty in predictions [20]. These methods show the potential of integrating advanced ensemble techniques within deep learning frameworks. This method shows the potential of integrating clustering-based techniques within deep learning frameworks.

B. Ensemble Pruning and Clustering-based Approaches

Ensemble pruning optimizes the selection of classifiers within an ensemble to balance efficiency and performance. Diverse strategies, including ranking-based methods and optimization-based methods like Optimum-Path Forest (OPF) ensemble pruning [21], illustrate the evolving landscape of ensemble pruning. Additionally, clustering-based approaches, which organize similar classifiers and select the most representative ones for the ensemble, add another dimension to enhancing the efficiency of ensemble learning systems [16].

C. Fuzzy Min-Max Neural Networks

Fuzzy Min-Max neural networks have become instrumental in classification and clustering. The advancements in these networks, such as the General Fuzzy Min-Max neural network and the Data-Core-based Fuzzy Min-Max neural network, have enhanced their ability to handle overlapping class distributions and improve classification accuracy. Recent studies propose models like the Deep Fuzzy Min-Max Neural Network (DFMNN), which introduce new initialization and optimization procedures to overcome input order and overlap region problems, significantly improving classification performance and robustness [22]. Another study utilized the Fuzzy Min-Max classifier for lung cancer subtyping from gene expression data, demonstrating its effectiveness in early cancer diagnosis by achieving high validation and cross-validation accuracies [23].

D. Dynamic Classifier Selection Techniques

Dynamic classifier selection techniques, focusing on estimating and utilizing the competence levels of classifiers, have seen significant advancements. Approaches like

META-DES and DSOC have emphasized the importance of the region of competence in classifier selection, leading to more accurate and tailored predictions. The refinement of these techniques, considering both correctly-classified and misclassified samples, has contributed to improved accuracy in dynamic selection systems, making them more effective in diverse machine learning applications [16].

E. Recent Advancements and Challenges

[24] exposed the shortcomings of traditional scoring systems like SAPS II, SOFA [25], and single machine learning models in critical care settings [26]. This work introduced the DELAK method, employing K-means sampling and dynamic ensemble selection to address these issues, an approach in line with current trends in ensemble methods for enhanced predictive performance [27], [28]. However, DELAK, like many other distance-based dynamic ensemble methods, struggled to capture the intricacies of complex data distributions and personalize predictions for new test instances, a limitation echoed in contemporary research [29], [30]. While DELAK's innovative fusion method via a distance-based dynamic ensemble was a step forward, it still faced challenges, especially in high-dimensional spaces, a topic explored in recent ensemble deep learning research [31]–[33].

Further contributions, such as the META-DES.Oracle framework in [34], extended the scope by utilizing meta-learning for classifier selection. While groundbreaking, this approach grappled with potential overfitting and computational complexities stemming from its intricate meta-feature framework and the optimization process using BPSO [35]. Additionally, its reliance on linear classifiers limited its effectiveness in scenarios with non-linear relationships. Recent ensemble learning advancements have focused on overcoming these limitations by exploring diverse methods like bagging, boosting, and decision tree ensembles, which have been shown to offer more robust solutions in handling complex data patterns [26]. These techniques, along with novel approaches such as ensemble methods for dirty data, provide a framework for integrating multiple learning algorithms and addressing the challenges observed in META-DES.Oracle, particularly in handling high-dimensional data and ensuring model generalization [36].

Similarly, [37] and [11] introduced novel selection methods, yet faced challenges in the sensitivity of unsupervised ensemble selection to the influence of class labels and the limitations of distance metrics for assessing classifier competence. These issues are also reflected in studies focusing on unsupervised data selection and ensemble classifiers, which further emphasize the complexities in selecting and combining multiple learning models without the guidance of labeled data [38]. Recent advancements in unsupervised ensemble classification, particularly in handling data dependencies and using clustering ensembles for feature selection, provide new perspectives and potential solutions to these challenges [39], [40], [41].

[42] tackled the issue of class overlapping and noise, highlighting the inadequacy of existing ensemble methods in handling these complexities. Additionally, [43] addressed the unique challenges in quantification tasks, proposing a dynamic ensemble method integrating data complexity measures. However, adaptability in dynamically changing data environments remained a challenge for these approaches, an issue further explored in recent works such as [44] dynamic neural network ensemble algorithm, which effectively balances diversity and accuracy in dynamic environments. New insights into dynamic ensemble methods for data streams are provided by [45], who discuss the integration of drift detection algorithms and dynamic updates, crucial for real-time data processing and [46], who review advanced ensemble techniques for time series forecasting, a critical aspect of dynamic environments.

Yet, it encountered difficulties in handling highly complex or high-dimensional data distributions, particularly in defining regions of competence and adapting to rapidly evolving data environments [47]. The challenges of adaptability in dynamically changing data environments are also addressed in the META-DES framework, which employs meta-learning techniques to enhance classifier competence through a variety of meta-features [45]. Additionally, the dynamic ensemble selection for quantification tasks, which further pushes the boundary of dynamic adaptability using data complexity measures, shows promising results in specific applications such as agricultural yield prediction and financial forecasting [43], [48].

Recent advancements in dynamic ensemble selection have addressed specific limitations of existing methods. [49] introduced the Fuzzy Hyperbox-based Dynamic Ensemble Selection (FH-DES) framework, recognizing the computational inefficiency and sensitivity to local data distributions inherent in KNN-based methods. FH-DES utilizes fuzzy hyperboxes for classifier competence estimation, offering a more efficient and nuanced approach. Additionally, their work introduces incremental learning capabilities and a novel Smooth Borders Membership function, further enhancing the accuracy and adaptability of the framework. However, challenges remain regarding potential sensitivity to local data distributions in complex datasets and the computational complexity of hyperbox management, particularly in large-scale problems.

III. Methodology

A. The proposed DDELA-GMM framework

This paper proposes a novel dynamic ensemble model that addresses the limitations identified in previous works, including those of FH-DES [49] and DELAK [24]. By employing Gaussian Mixture Models (GMM) for clustering and a density-based decision-making approach, our model offers a more granular analysis of complex data distributions and a more robust framework for classifier competence estimation. This shift from distance-based to density-based methods, combined with a dynamic weighting mechanism, promises improved adaptability, accuracy,

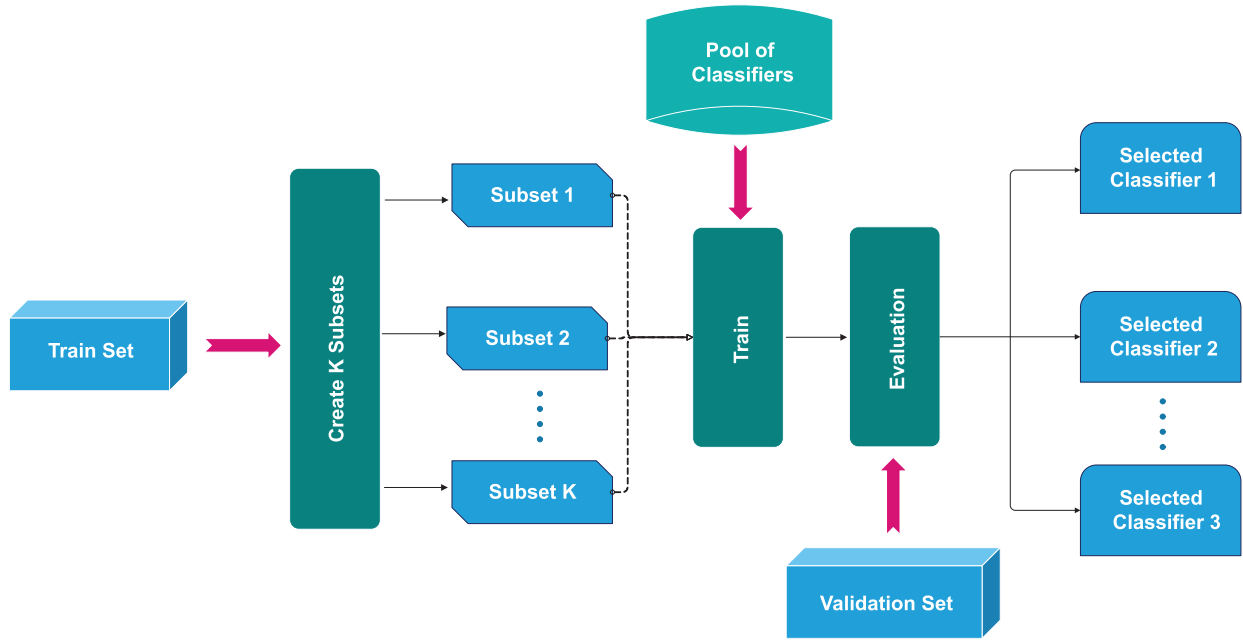


Fig. 1. Training phase of proposed model

and efficiency in handling complex and high-dimensional datasets. As illustrated in Figure 1, the training phase of our proposed model involves leveraging GMM to effectively capture the underlying structure of the data. This approach facilitates the creation of specialized classifiers that are adept at recognizing specific data patterns within complex and overlapping distributions. In comparison to FH-DES, which suffers from sensitivity to local data distributions due to its reliance on hyperboxes for clustering, our use of GMMs allows for more sophisticated modeling of overlapping and multi-modal distributions. GMMs, with their capability to adapt cluster shapes based on the data, provide a flexible and nuanced understanding of data distributions, enabling more accurate identification of competence regions for classifiers.

Furthermore, as depicted in Figure 2, the generalization phase employs a density-based dynamic ensemble selection mechanism. This ensures that, during prediction, the most relevant classifiers contribute more significantly based on the density estimation of the input data point, thereby enhancing the model's overall performance and robustness.

Moreover, when compared to DELAK, our model surpasses the limitations of K-means clustering and distance-based weighting by introducing GMM-based clustering and density-based classifier selection. This approach not only handles complex and overlapping data distributions more effectively but also improves adaptability to dynamic environments.

Significant contributions in the realm of GMM for ensemble learning include [50]'s work on Gaussian mixture learning, which improves the efficiency and accuracy of ensemble models in dynamic environments by combining hierarchical clustering with the Expectation-Maximization algorithm. Additionally, [51]'s recent work on robustly

learning mixtures of Gaussians provides theoretical backing for the efficacy of GMM in complex distribution scenarios, supporting our model's approach. This approach allows our model to excel in scenarios where data patterns are intricate and where adaptability to changing data distributions is crucial. By effectively capturing the nuances of complex data and dynamically weighting the contributions of individual classifiers, our model offers a significant advancement in the field of ensemble learning.

We introduce an advanced ensemble learning strategy that is deeply rooted in the principles of data density and specialization. Our proposed methodology consists of two interrelated components: the Gaussian Mixture Model (GMM) Sampling and the Density-based Dynamic Ensemble.

B. Gaussian Mixture Model (GMM) Clustering & Sampling

Traditional ensemble techniques often create subsets of data either randomly or using simple heuristics. While effective for many datasets, the increasing complexity and multimodal nature of modern datasets necessitate a more sophisticated approach. Enter Gaussian Mixture Model (GMM) Sampling, a method designed to capture the intricate structures within datasets and thereby enhance the versatility of ensemble models.

Motivation: The driving force behind employing GMM in the ensemble paradigm is its ability to recognize and capture underlying patterns within datasets. Given that many datasets exhibit a multimodal distribution, GMM's capability to model data as a mixture of several Gaussian distributions becomes invaluable. By ensuring that each base classifier in the ensemble is trained on data that mirrors these Gaussian components, we bolster

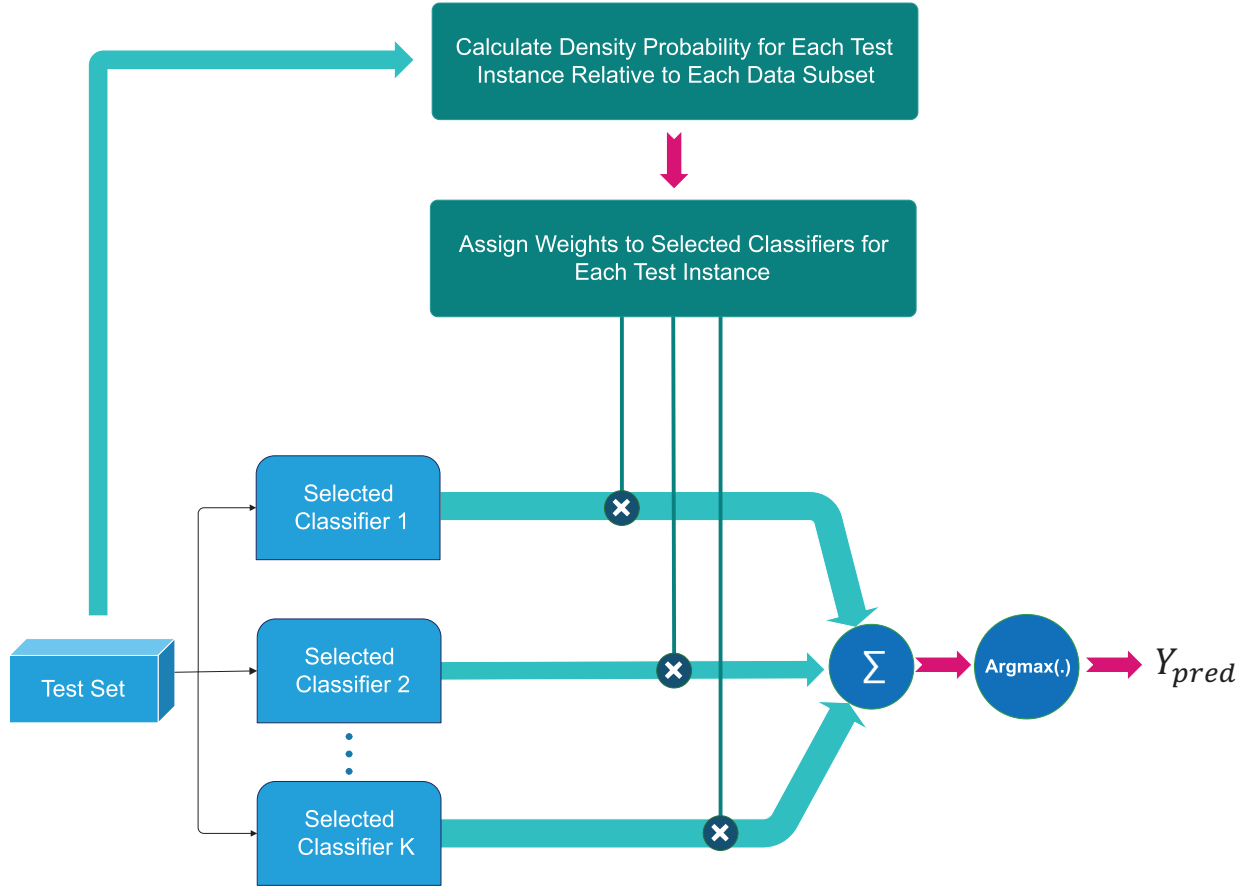


Fig. 2. Generalization phase of proposed model

the ensemble's capacity to recognize and respond to subtle data structures.

GMM Overview: Mathematically, a Gaussian Mixture Model represents data as a weighted combination of M Gaussian distributions as shown in Equation 1.

$$p(\mathbf{x}) = \sum_{i=1}^M \pi_i \mathcal{N}(\mathbf{x} | \mu_i, \Sigma_i) \quad (1)$$

Where:

- x is a data point.
- π_i denotes the mixture weight of the i^{th} Gaussian.
- $\mathcal{N}(\mathbf{x} | \mu_i, \Sigma_i)$ is the Gaussian distribution with parameters mean μ_i and covariance Σ_i .

The Gaussian distribution $\mathcal{N}(\mathbf{x} | \mu_i, \Sigma_i)$ is defined as shown in Equation 2:

$$\mathcal{N}(\mathbf{x} | \mu_i, \Sigma_i) = \frac{1}{(2\pi)^{d/2} |\Sigma_i|^{1/2}} \times \exp\left(-\frac{1}{2} (\mathbf{x} - \mu_i)^T \Sigma_i^{-1} (\mathbf{x} - \mu_i)\right) \quad (2)$$

where d is the dimensionality of x .

Parameter Estimation via Expectation-Maximization (E-M): The crux of GMM lies in the accurate estimation of its parameters. The E-M algorithm serves this purpose.

It's an iterative algorithm, progressively refining the parameters to maximize the likelihood of observing the given data:

- 1) **E-Step:** Computes the posterior probabilities, essentially estimating the likelihood of a data point originating from a specific Gaussian component.

$$\gamma_{ji} = \frac{\pi_i \mathcal{N}(x_j | \mu_i, \Sigma_i)}{\sum_{k=1}^K \pi_k \mathcal{N}(x_j | \mu_k, \Sigma_k)} \quad \text{for } i = 1, 2, \dots, K \quad (3)$$

- 2) **M-Step:** Refines the parameters based on the probabilities obtained from the E-Step, ensuring that the overall likelihood of the data given the model parameters is maximized.

The objective function Q that the E-M algorithm aims to maximize is given by Equation 4.

$$Q(\theta | \theta^{(t)}) = \sum_{i=1}^N \sum_{j=1}^M \gamma_{ij}^{(t)} [\log \pi_j + \log \mathcal{N}(\mathbf{x}_i | \mu_j, \Sigma_j)] \quad (4)$$

where $\gamma_{ij}^{(t)}$ is the posterior probability of the i^{th} data point belonging to the j^{th} Gaussian component at iteration t .

The update equations for the M-step are in Equations 5 to 7.

$$\pi_j^{(t+1)} = \frac{1}{N} \sum_{i=1}^N \gamma_{ij}^{(t)} \quad (5)$$

$$\mu_j^{(t+1)} = \frac{\sum_{i=1}^N \gamma_{ij}^{(t)} \mathbf{x}_i}{\sum_{i=1}^N \gamma_{ij}^{(t)}} \quad (6)$$

$$\Sigma_j^{(t+1)} = \frac{\sum_{i=1}^N \gamma_{ij}^{(t)} (\mathbf{x}_i - \mu_j^{(t+1)}) (\mathbf{x}_i - \mu_j^{(t+1)})^T}{\sum_{i=1}^N \gamma_{ij}^{(t)}} \quad (7)$$

Implications of GMM Sampling in Ensemble Learning: Once the GMM parameters are derived, they play a pivotal role in segmenting the data. Each Gaussian component identified by the GMM provides insights into a specific data structure or pattern. When data points are assigned to subsets based on their association with these Gaussian components, it ensures that each subset is representative of a distinct data pattern. Training individual classifiers on these subsets equips the ensemble with a diverse set of experts, each attuned to a specific data structure. This nuanced approach to data sampling offers several advantages:

- **Diversity:** Ensures that base classifiers capture diverse perspectives of the data.
- **Specialization:** Each base classifier becomes an expert in a specific data pattern, enhancing the ensemble's overall competence.
- **Robustness:** The ensemble becomes more resilient to noise and anomalies, given that the GMM-based segmentation inherently reduces the impact of outliers on any individual classifier.

Operational Steps: With the GMM parameters in place, the methodology involves ranking data points based on their likelihoods under each Gaussian component. A sampling ratio, ϕ , then determines the fraction of top-ranking data points to be included in each subset. These subsets, rich in structure and pattern, serve as the training data for individual base classifiers. The detailed algorithmic steps for this process are outlined in Algorithm 1.

C. Density-based Dynamic Ensemble Algorithm

Traditional ensemble learning methods often rely on the simple aggregation of base classifiers' outputs, without considering the inherent nuances associated with the data regions that different base classifiers might be specialized in. The Density-based Dynamic Ensemble methodology emerges as an innovative response to this limitation, ensuring that the ensemble's predictions are sensitive to the intrinsic data densities associated with test instances.

Motivation: In fact, the base classifiers are in competition, and their contribution to the final decision depends on the density probability of the test instance relative to the data subset each classifier was trained on. Given that different data subsets (derived from Gaussian Mixture Model (GMM) sampling) capture unique data structures,

Algorithm 1 GMM Sampling

Input: Sampling ratio ϕ ; Training dataset D ; Training data without labels X ; Cluster number K

Output: Data subsets $D' = D'_1 \cup D'_2 \cup \dots \cup D'_K$

- 1: procedure GMM Sampling(ϕ, X, K)
 - 2: Initialize K Gaussian components with means
 - 3: $\mu_1, \mu_2, \dots, \mu_K$ and covariances $\Sigma_1, \Sigma_2, \dots, \Sigma_K$
 - 4: Initialize mixing coefficients $\pi_1, \pi_2, \dots, \pi_K$ such that $\sum_{i=1}^K \pi_i = 1$
 - 5: repeat
 - 6: E-step: Compute the responsibilities for each sample $x_j \in X$ using Equation 3.
 - 7: M-step: Update μ_m, Σ_m, π_m using Equations 5, 6, and 7.
 - 8: until convergence of parameters μ_i, Σ_i , and π_i
 - 9: for $i = 1, 2, \dots, K$ do
 - 10: Compute the responsibilities for each instance in X with respect to Gaussian component i
 - 11: Select $\text{int}(m \times \phi)$ instances with the highest responsibilities into partition C'_i
 - 12: end for
 - 13: Return the labels in D to instances in C' to get K data subsets $D' = D'_1 \cup D'_2 \cup \dots \cup D'_K$
 - 14: end procedure
-

the classifiers trained on these subsets specialize in different regions of the data distribution. Therefore, for any given test instance, it is essential to evaluate the competence of each classifier based on the density of the test instance within the relevant data subset, allowing the ensemble to leverage the classifiers with the highest competence.

Density Estimation: The density of a test instance with respect to a data subset is indicative of the likelihood that the test instance belongs to that subset. Mathematically, the density $\rho_i(z)$ of a test instance z concerning a subset D'_i is calculated using the Gaussian density function as shown in Equation 8:

$$\rho_i(z) = \frac{1}{(2\pi)^{\frac{d}{2}} |\Sigma_i|^{\frac{1}{2}}} \exp \left(-\frac{1}{2} (z - \mu_i)^T \Sigma_i^{-1} (z - \mu_i) \right) \quad (8)$$

Where μ_i and Σ_i are the mean and covariance matrix, respectively, of the Gaussian component associated with subset D'_i . **Classifier Invocation:** After computing densities, the weight of each classifier is determined based on the density of the test instance with respect to the subset the classifier was trained on. This means that all classifiers contribute to the final prediction, but their influence is modulated by how well the test instance aligns with the data distribution of the corresponding subset. The weights are proportional to the densities, giving more importance to classifiers associated with higher densities for the test instance.

Decision Aggregation: An ensemble's strength lies in its ability to amalgamate insights from multiple base classifiers. In the Density-based Dynamic Ensemble approach,

predictions from all base classifiers are aggregated using a weighted voting scheme. Specifically, the probability vector for each class is calculated by weighting the predictions of each classifier according to the density of the test instance with respect to the subset on which that classifier was trained. The final predicted class label is the one with the highest combined probability. The algorithmic steps for this methodology are presented in Algorithm 2.

$$y_{\text{pred}} = \operatorname{argmax}(\mathbf{y}_{\text{prob}}) \quad (9)$$

Where $\mathbf{y}_{\text{prob}}$ is the probability vector of the test instance belonging to each class.

Algorithm 2 The density-based dynamic ensemble for multi-class classification.

Require: Training dataset $D' = D'_1 \cup D'_2 \cup \dots \cup D'_K$; Test example $e_i = (e_i^1, e_i^2, \dots, e_i^n)^T$; Gaussian components $\mu = \{\mu_i | i = 1, 2, \dots, K\}$ and $\Sigma = \{\Sigma_i | i = 1, 2, \dots, K\}$.
Ensure: Predicted class label y_{pred} ; Predicted probability vector $\mathbf{y}_{\text{prob}}$.

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1: procedure Base Classifiers Training( $D'$ )
2:   for  $\eta = 1, 2, \dots, K$  do
3:     Train base model  $h_\eta(\cdot)$  on training data subset  $D'_\eta$ 
4:   end for
5:   Return  $h = \{h_\eta(\cdot) | \eta = 1, 2, \dots, K\}$ 
6: end procedure
7: procedure Ensemble( $e_i, \mu, \Sigma, h$ )
8:   Use each base classifier to predict the label of  $e_i$  to get a result vector  $\mathbf{h}(e_i) = (h_1(e_i), h_2(e_i), \dots, h_K(e_i))^T$ 
9:   Compute the density of test example  $e_i$  with respect to each subset  $D'_i$  using the Gaussian density function:
10:  
$$\rho_i(e_i) = \frac{1}{(2\pi)^{\frac{d}{2}} |\Sigma_i|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(e_i - \mu_i)^T \Sigma_i^{-1} (e_i - \mu_i)\right)$$

11:   Compute the weight vector for test example  $e_i$ :
12:  
$$\mathbf{w}_i = \left( \frac{\rho_1(e_i)}{\sum_{j=1}^K \rho_j(e_i)}, \frac{\rho_2(e_i)}{\sum_{j=1}^K \rho_j(e_i)}, \dots, \frac{\rho_K(e_i)}{\sum_{j=1}^K \rho_j(e_i)} \right)^T$$

13:   Compute the probability vector of  $e_i$  belonging to each class:  $\mathbf{y}_{\text{prob}} = \mathbf{w}_i^T \cdot \mathbf{h}(e_i)$ 
14:   Assign the predicted class label:
15:    $y_{\text{pred}} = \operatorname{argmax}(\mathbf{y}_{\text{prob}})$ 
16: end procedure
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Implications: The Density-based Dynamic Ensemble approach offers several advantages:

- **Precision:** By considering data densities, the methodology ensures that all classifiers are considered, but the most relevant ones are given greater weight, enhancing classification precision.
- **Adaptability:** The approach is inherently adaptable, adjusting the ensemble's response based on the test instance's location within the data distribution.
- **Reduced Overfitting:** Leveraging density information ensures that the ensemble doesn't overly rely on any single classifier, reducing the chances of overfitting.

In summation, the Density-based Dynamic Ensemble methodology infuses ensemble learning with a level of sensitivity and adaptability, ensuring that predictions are both accurate and attuned to the intricacies of the dataset's distribution.

IV. Results and Discussion

The performance of various ensemble methods, with a base learner count $nc = 6$ and a regularization parameter $\phi = 0.5$, is systematically compared in Table 1. This table assesses four ensemble methods—DensityE, DistE, AvgE, and MaxE—using four distinct types of base learners: logistic regression (lr), support vector machine (svm), k-nearest neighbors (knn), and decision tree (dt).

A. Ensemble Method Performance Analysis

Table 1 reveals that the DensityE ensemble method consistently achieves the highest mean scores across most base learners. Specifically, the logistic regression base learner combined with the DensityE method attains a mean score of 0.973, closely followed by the support vector machine base learner with a mean score of 0.969. The k-nearest neighbors base learner also performs well under the DensityE method, achieving a mean score of 0.958. Conversely, the decision tree base learner shows significantly lower performance across all ensemble methods, with the highest mean score of 0.841 observed under the DistE method.

The DistE method also exhibits strong performance, particularly with logistic regression and support vector machine base learners, achieving mean scores of 0.962 and 0.959, respectively. The AvgE and MaxE methods, however, generally demonstrate lower performance across all base learners. The AvgE method performs moderately well with support vector machine and k-nearest neighbors, achieving mean scores of 0.897 and 0.856, respectively. The MaxE method shows the least effective performance, especially with the logistic regression base learner, which only achieves a mean score of 0.508.

B. Impact of Cluster Number and Sampling Ratio

Figure 4 illustrates the relationship between the number of clusters and the F-score for the DensityE ensemble method. The support vector machine consistently outperforms other base learners across various cluster numbers, maintaining a high F-score close to 0.97. Logistic regression follows closely, while k-nearest neighbors and decision tree show lower performance. This trend suggests that increasing the number of clusters generally enhances the F-score, indicating that more detailed partitioning of the data can lead to better ensemble performance. However, the improvement plateaus beyond a certain number of clusters, suggesting a point of diminishing returns.

Figure 3 presents the F-score plotted against the sampling ratio for the DensityE ensemble method. As the sampling ratio increases, the F-score improves for all base learners. The support vector machine achieves the highest F-score, with logistic regression and k-nearest neighbors

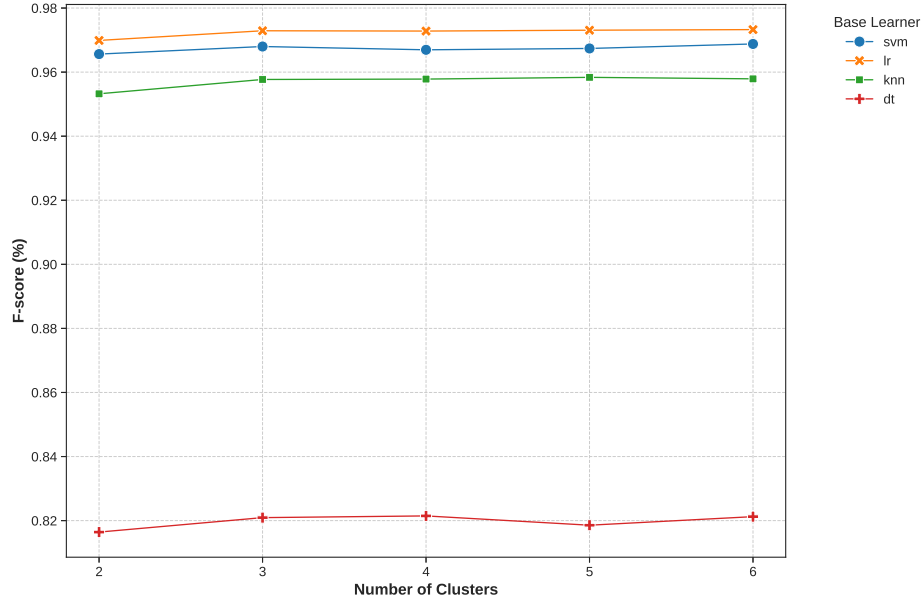


Fig. 3. F-score vs Number of Clusters for Density ensemble method

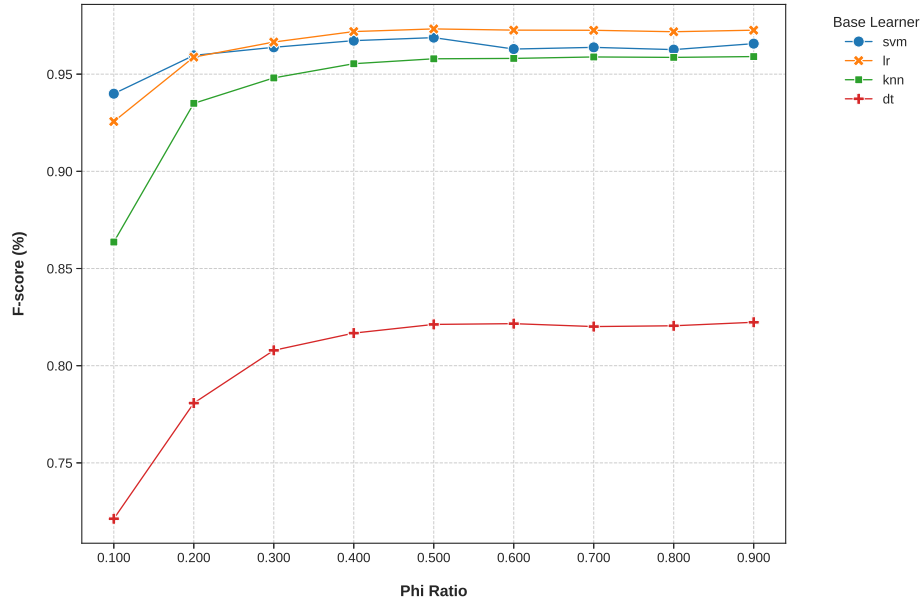


Fig. 4. F-score vs Sampling Ratio for Density ensemble method

following. The decision tree base learner, once again, shows the lowest performance. This positive correlation between sampling ratio and F-score highlights the importance of using a larger sample size to enhance model performance, likely because larger samples provide more representative data, allowing the ensemble to make more accurate predictions.

Figure 5 provides a visual representation of the distribution of scores for each ensemble method and base learner combination. This figure reinforces the tabulated results, highlighting the overall superior performance of

the DensityE and DistE methods. The variability in scores is also depicted, with DensityE and DistE showing tighter distributions compared to AvgE and MaxE. Notably, the single SVM model achieves a high and consistent performance, underscoring the robustness of SVM as a standalone model.

V. Discussion

The results unequivocally indicate that the DensityE ensemble method offers superior performance compared to other ensemble methods, particularly when logistic

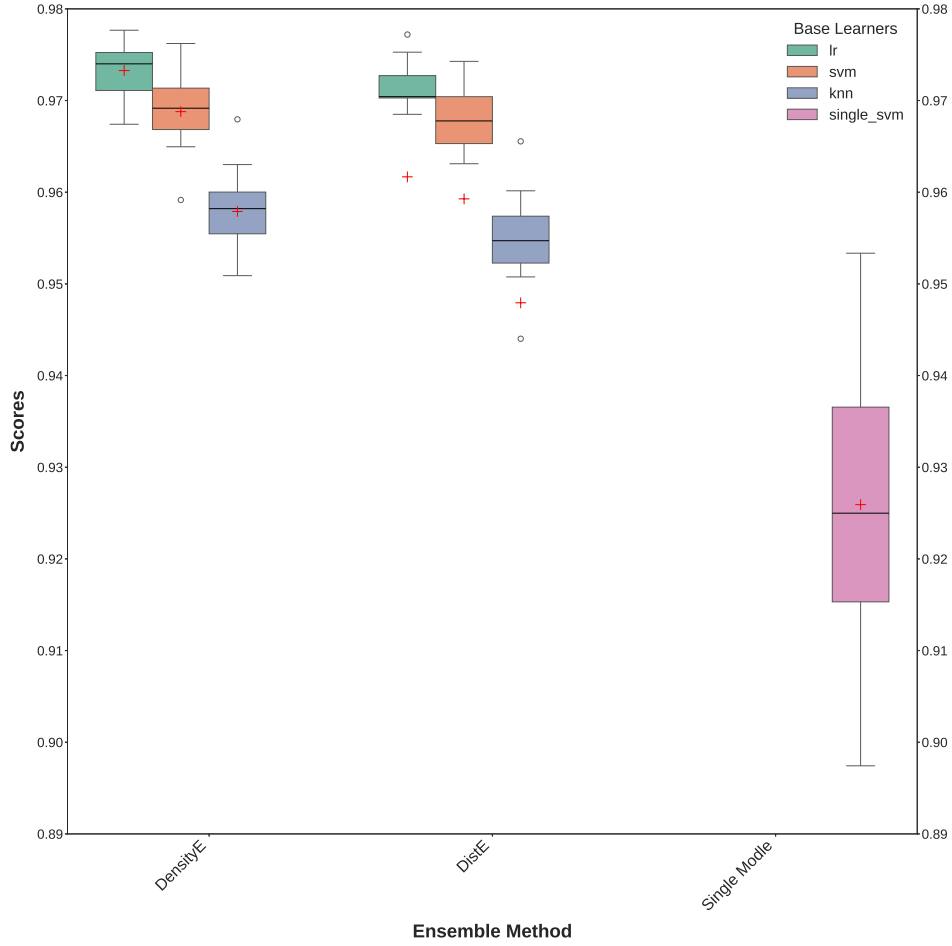


Fig. 5. Comparison of F-score Distributions Across Different Ensemble Methods and Base Learners

TABLE I
Comparison of Ensemble Methods with Different Base Learners

Ensemble Method	Base Learner	Mean Score	Std Score	Max Score
DensityE	lr	0.973	0.003	0.978
	svm	0.969	0.003	0.976
	knn	0.958	0.004	0.968
	dt	0.821	0.010	0.839
DistE	lr	0.962	0.043	0.977
	svm	0.959	0.040	0.974
	knn	0.948	0.031	0.966
	dt	0.841	0.016	0.864
AvgE	lr	0.750	0.210	0.965
	svm	0.897	0.115	0.965
	knn	0.856	0.118	0.951
	dt	0.720	0.108	0.827
MaxE	lr	0.508	0.034	0.574
	svm	0.855	0.040	0.907
	knn	0.722	0.122	0.892
	dt	0.410	0.143	0.653

regression and support vector machine are utilized as base learners. This finding underscores the efficacy of DensityE's approach in leveraging the strengths of diverse base learners and effectively managing their combined output. The superior performance of logistic regression and support vector machine base learners can be attributed to their robustness and generalization capabilities in high-

dimensional spaces, which are prevalent in the datasets employed in this study.

The comparative analysis across different base learners highlights the critical role of base learner selection in ensemble methods. While linear models like logistic regression and support vector machine benefit significantly from the ensemble approach, non-linear models such as k-

nearest neighbors and decision trees show relatively lower performance. This disparity may be due to the inherent differences in how these models handle data complexity and variability.

The insights gained from Figures 1 and 2 provide a deeper understanding of ensemble dynamics. Increasing the number of clusters generally leads to better performance, but the benefits diminish beyond a certain point. Similarly, a higher sampling ratio enhances the F-score, emphasizing the value of larger training samples for achieving higher accuracy.

In conclusion, the DensityE ensemble method, particularly when paired with logistic regression and support vector machine base learners, demonstrates the most promising performance. Future research could explore the integration of more sophisticated base learners and the application of these ensemble methods to various types of datasets to further validate their robustness and generalizability. Additionally, investigating the effects of varying other parameters within the DensityE method may yield further performance improvements and insights.

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